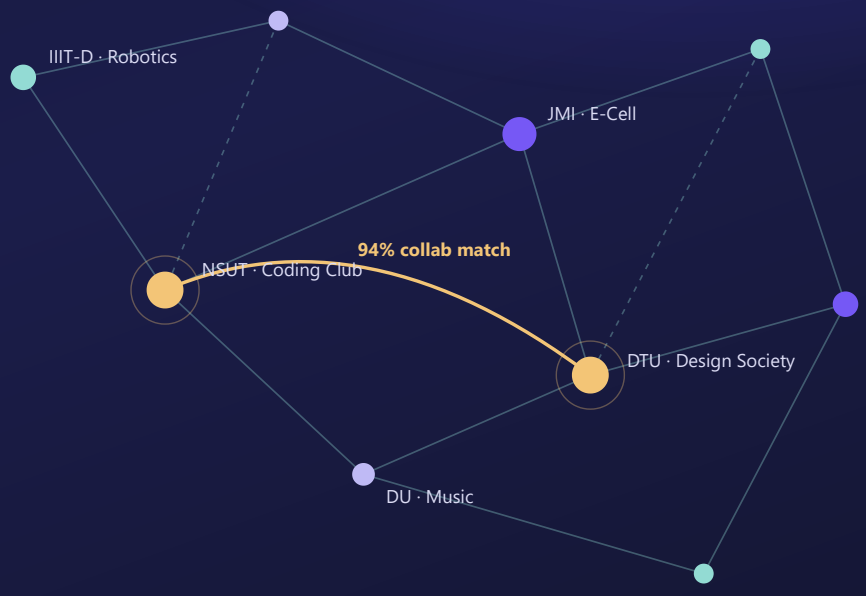
PROBLEM STATEMENT 2 · PRODUCT & ARCHITECTURE BLUEPRINT

CampusConnect

Connect beyond campus.

A verified, AI-assisted network where colleges, student communities and students **discover** each other, **promote** opportunities across campuses and **run collaborations end-to-end**, from finding the right partner to handing the relationship over to next year's team.

12 product innovations | Event-driven, cloud-native | Built for 1M daily users | India-first · DPDP-aligned | Web + iOS + Android



Submitted by **Team [Your Team Name]**
[College Name]

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At a glance

4.3 Cr
students in Indian higher education (AISHE 2021-22). Their communities are almost entirely disconnected from each other.

45,000+
colleges, each running dozens of clubs, fests and chapters inside a single campus.

12
original features covering discovery, promotion, collaboration and continuity.

1M DAU
design target for Year 3 across 3,000+ colleges, sized and capacity-planned in §15.

99.9%
availability SLO, with p99 < 1 s message delivery and RPO under 1 minute.

Day 1
value for a single club with zero network: our answer to the cold-start problem.

EXECUTIVE SUMMARY

01 An operating layer for *inter-college* student life

CampusConnect turns thousands of isolated campus communities into one verified, searchable and collaborative network, and gives every community the tools to actually *do* something together once they've found each other.

The problem (PS 2)

Student communities operate in isolation. Finding relevant organizations at other colleges is hard. Promotion stops at the campus gate, inside single-campus WhatsApp groups. Collaborations depend on whoever has personal contacts this year.

Our core insight

Isolation isn't caused by a missing feed. Three systems are missing: **(1)** a way to find a *complementary* partner, **(2)** a low-friction way to plan and run things together, and **(3)** institutional memory that survives the annual leadership turnover. Generic social networks offer none of the three.

The solution

Five pillars (**Discover · Connect · Promote · Collaborate · Sustain**) delivered through 12 features, including AI partner matching (**Collab Compass**), clash-aware scheduling (**Date Doctor**), poster-to-event publishing (**Event Launchpad**), shared co-host workspaces (**Collab Rooms**) and leadership-handover memory (**Legacy Vault**).

Built to scale

A modular monolith that grows into event-driven services. PostgreSQL is the source of truth, and search, feeds and analytics are rebuildable projections. It runs on AWS Mumbai with multi-AZ redundancy and a Hyderabad DR region. Capacity-planned for 1M DAU and spike-safe for fest registration rushes.

How we beat the cold start. A social network is useless with no users, but CampusConnect isn't only a social network. A single club gets value on day 1: it publishes events from a poster in 30 seconds, syndicates them to WhatsApp and Instagram, checks dates against public calendars and keeps its Legacy Vault. Each of those actions adds a node to the network, and every shared link and collab request invites the next one.

What judges should remember

Complete PS coverage

Every sentence of PS 2 maps to a named feature and a metric (§03).

Novel, not a clone

Complementarity matching, clash detection, resource exchange and leadership-handover memory don't exist together in any student platform we know of.

Production-grade engineering

Outbox pattern, CDC, cell-based scale path, SLOs, DR, DPDP compliance, all specified in detail.

India-first

WhatsApp-first distribution, Hindi + regional languages, Lite mode for low-end phones, data stored in India.

Institution value

College HQ gives approvals, safety oversight and collaboration records useful for accreditation documentation.

A business

Free for students. Revenue from College HQ licences, a Sponsor Pool, community Pro plans and ticketing fees.

THE PROBLEM, EXAMINED IN DEPTH

02 Why campus communities stay *islands*

We broke PS 2 into its four stated failures, then went one level deeper to find the root causes a generic platform would miss.

PAIN (FROM PS 2)	WHAT HAPPENS TODAY	ROOT CAUSE	CAMPUSCONNECT ANSWER
Operating in isolation	Clubs know peers at other colleges only through seniors' personal contacts.	No shared, persistent graph across campuses.	Constellation graph, verified directory, cross-campus channels.
Discovering relevant organizations	Searching Instagram handles; asking around on WhatsApp.	Community data is unstructured and unsearchable.	Semantic + geo discovery, Campus Pulse map, Ask CampusConnect.
Promoting events	Posters forwarded inside own-campus groups; reach stops at the gate.	Distribution is siloed with no audience targeting.	Event Launchpad, reach targeting, Syndication Bridge, Smart Digest.
Establishing collaborations	Cold DMs go unanswered; co-hosting is coordinated in chaotic group chats.	No matching, no trust signals, no shared workspace.	Collab Compass, Collab Credits, Collab Rooms, Resource Exchange.

Hidden problems we uncovered

Annual leadership churn

Office-bearers change every academic year. Sponsor contacts, partner relationships and playbooks leave with them. → **Legacy Vault**

Calendar collisions

Nearby colleges schedule fests and hackathons on the same weekend and split the audience, and exam weeks kill turnout. → **Date Doctor**

Resource asymmetry

One campus has a 500-seat auditorium, another has 300 developers, a third has industry judges. → **Resource Exchange**

Trust deficit

"Is this club real? Will they deliver?" There's no reputation that travels across campuses. → **Verification tiers + Collab Credits**

Institutional blind spot

Colleges can't see, approve or report what their communities do across campuses. → **College HQ**

Wrong channel

Students live on WhatsApp and Instagram and won't check yet another app daily. → **Syndication Bridge + WhatsApp digests**

Who we build for

Student

Aarav, 2nd-year B.Tech

"I want robotics clubs and events beyond my campus, and I need to know they're legit."

Needs: interest + distance discovery, trust, one-tap registration.

Community lead

Priya, coding club president

"We need judges, a 400-seat venue and a design partner for our hackathon."

Needs: partners, reach beyond campus, coordination, sponsors.

College

Dr. Mehta, Dean of Student Affairs

"I approve 60 events a year and can't track any of them afterwards."

Needs: approvals, safety, visibility, reports.

Sponsor

Rohan, startup community manager

"I want to reach 20 colleges' developer clubs without sending 20 cold emails."

Needs: targeted reach, measurable outcomes.

SOLUTION

03 Five pillars, one *verified graph*

Each pillar answers one verb from the problem statement. Together they turn "we found each other" into "we did something together, and next year's team can repeat it".



Figure 1 · Product pillars on top of a single verified graph.

Problem-statement coverage matrix

PS 2 REQUIREMENT	CAMPUSCONNECT CAPABILITY	PROOF METRIC
"Communities operate in isolation"	Constellation graph, cross-campus channels, proactive Collab Compass suggestions	% of communities with ≥ 1 cross-campus connection
"Difficult to discover relevant organizations"	Hybrid semantic + geo search, Campus Pulse map, Ask CampusConnect, interest onboarding	Search → follow conversion; time-to-first-follow
"Promote events"	Event Launchpad, campus/city/national reach targeting, Syndication Bridge, Smart Digest, Date Doctor	Share of registrations from other colleges
"Establish collaborations"	Collab Compass, structured collab requests, Collab Rooms + Charter, Resource Exchange, Collab Credits	**North Star:** cross-campus collaborations completed / month
"Enables colleges … to connect"	College HQ, institution verification, collaboration & MoU registry, approvals	Colleges with an active HQ; approvals SLA
"Promote opportunities across campuses"	Opportunity Board (judges, speakers, volunteers, internships), targeted reach, Campus Passport	Applications per opportunity; fill rate

Design principle: "Bharat-first". English + Hindi at launch, then Tamil, Telugu, Bengali and Marathi. A Lite mode for 3G and entry-level Android phones. WhatsApp-first notifications, and every public page readable without logging in. Accessibility to WCAG 2.2 AA.

TWELVE INNOVATIONS

04 Features that don't exist *anywhere else*

Each one targets a root cause from §02 and has a concrete implementation path, so none of it is hand-waving.

Collab Compass 01

AI partner matching built on complementarity, not similarity.

PROBLEM Partners are found through personal contacts, and cold DMs go unanswered.

HOW Every community has a *skills* profile (what it's good at) and a *goals* profile (what it wants to do this term). Compass recommends partners whose skills complement yours, whose goals overlap, who are nearby and who are trusted, and it shows *why*: "They bring UI/UX and branding; you bring ML. Both plan a November hackathon. 6 km apart."

pgvector embeddings · explainable match score (§12) · weights learned from completed collaborations

Date Doctor 02

Scheduling that knows what clashes with your event.

PROBLEM Nearby colleges schedule big events on the same weekend and split the audience.

HOW When picking a date, organizers see a heat strip of overlapping events *for the same audience* within a radius, plus known exam windows and festivals. They get the three best slots, each with a reason.

PostGIS radius × audience-overlap (interest embeddings) · academic calendars from College HQ · per-day clash score

Event Launchpad 03

From poster to live event in 30 seconds.

PROBLEM Clubs already make a poster. Retyping details into every platform is what kills promotion.

HOW Drop a poster, Instagram link or WhatsApp text. AI extracts the title, date, venue, fee, eligibility and registration link, and highlights low-confidence fields for confirmation. Publishing auto-generates an OG card, a QR code and bilingual descriptions.

multimodal LLM with JSON-schema output · per-field confidence · geocoding · human-in-the-loop confirmation

Syndication Bridge 04

Reach students where they already are.

PROBLEM Indian students live on WhatsApp and Instagram, not on yet another app.

HOW Publish once, then push to WhatsApp Channels, Instagram, LinkedIn and email digests. Every share carries a tracked short link, so organizers see *which college and channel* drove registrations.

WhatsApp Cloud API + Meta Graph connectors · short-link service · attribution in ClickHouse

Collab Room + Charter 05

A shared workspace the moment two clubs say yes.

PROBLEM Co-hosting runs in chaotic group chats, and responsibilities and money get lost.

HOW Accepting a collab request auto-creates a room with a task board, cross-org roles, a budget-split ledger, files and a joint registration page. It also produces a one-page **COLLAB CHARTER** (who does what, logo placement, revenue split) that both sides approve.

real-time sync over WebSockets · charter rendered to PDF and stored as a signed, versioned record

Resource Exchange 06

Share what you have, borrow what you lack.

PROBLEM Resource asymmetry: one campus has an auditorium, another has developers, judges or 3D printers.

HOW Communities list lendable venues, equipment and people (judges, speakers, mentors, volunteers) with availability. Others request bookings, and completed exchanges earn Collab Credits.

Postgres `tstzrange` slots + an `EXCLUDE` constraint, so double-booking is impossible at the database level

Collab Credits & Trust Score 07

A reputation that travels across campuses.

PROBLEM "Will this club actually deliver?"

HOW After each collaboration, both sides leave a structured review (delivery, communication, turnout). Combined with the verification tier and history, this produces a Trust Score on the profile. Credits earned by helping others unlock featured placement in Discover.

Bayesian-average ratings (no small-sample bias) · decay across terms · graph anomaly detection against review rings

Legacy Vault 08

Club memory that survives graduation.

PROBLEM Every year the entire leadership graduates, and contacts, playbooks and partner relationships go with them.

HOW A structured handover: a role-succession workflow, a vault of documents, a partner/sponsor CRM and past-event playbooks. New leads can ask **"ASK THE VAULT"** things like "How did we book the auditorium last year?" and get answers with citations.

RAG over vault docs (pgvector + LLM) · `term` column on memberships · permission transfer on approved succession

Campus Pulse 09

The city's student life, live on a map.

PROBLEM Students can't see what's happening around them across campuses.

HOW A map and heat layer of events, open calls and active communities in a chosen radius and time window ("this weekend, within 8 km"), with travel time from your campus.

PostGIS + vector tiles (MapLibre) · server-side clustering · tiles cached at the edge · routing API for travel time

Campus Passport 10

One identity for every campus.

PROBLEM Every event uses a different form, and proof of participation is scattered.

HOW One QR identity for registration and check-in at any college. Check-ins add stamps and badges, and organizers issue **VERIFIABLE CERTIFICATES** that anyone can validate from a public link, ready to share on LinkedIn.

signed short-lived QR tokens (verifiable offline by the organizer app) · certificates as signed credentials (W3C VC-inspired)

Ask CampusConnect 11

Talk to the graph in plain language.

PROBLEM Filters can't express "robotics clubs in Delhi that co-hosted a hackathon and have 200+ members".

HOW A natural-language assistant that turns questions into safe, structured queries over the public graph and returns *cards*, not paragraphs.

LLM tool-calling against a whitelisted query API (never raw SQL) · permission-aware results · cached common intents

College HQ & Sponsor Pool 12

Value for institutions and the people who fund events.

PROBLEM Colleges can't see what their communities do. Small clubs can't attract sponsors alone.

HOW HQ gives institutions approvals with risk flags, safety oversight, a collaboration/MoU registry and exportable activity reports. Sponsor Pool bundles several clubs' audiences into one sponsorship deal with a transparent split.

role-scoped dashboards · k-anonymity (no segment under 50 people is ever shown to sponsors) · report exports

Bonus: Smart Digest. A weekly, personalized "what's worth your weekend" digest delivered on WhatsApp or email, ranked by the same engine as the feed. It brings students back without asking them to open the app every day.

PRODUCT DESIGN

05 Key screens, designed for *action*

Every core screen ends in one obvious action. Explanations ("why this match", "why this date") build trust in the AI instead of hiding it.

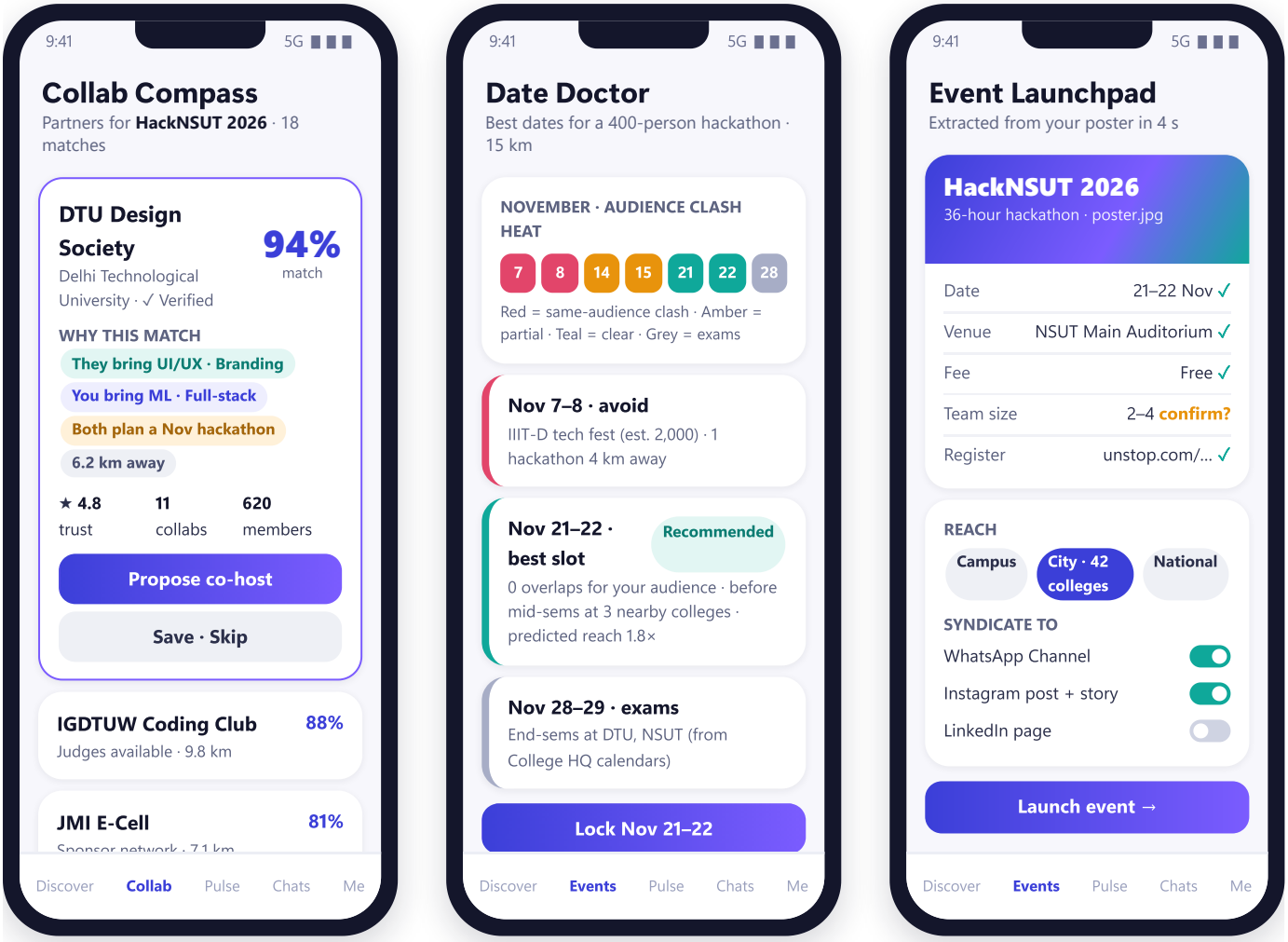


Figure 2 · Collab Compass, Date Doctor and Event Launchpad (mobile). Data shown is illustrative.

Design system

Tokens, not hex codes

Colour, type, spacing and radius live as shared tokens in `packages/ui` and drive both Tailwind (web) and NativeWind (mobile). Light and dark themes come free, and each college can theme its own HQ.

Explainable AI patterns

Every AI output shows its reasons as chips, a confidence state and an "edit" path. Nothing AI-generated gets published without a human confirming it.

Accessible & multilingual

WCAG 2.2 AA contrast, dynamic type, screen-reader labels and reduced motion. ICU message format with Devanagari-safe typography, plus Hinglish-tolerant search.

USER JOURNEYS

06 Four journeys, *end to end*

These are the flows we optimise and instrument first. Each step maps to a service in §08.

J1 · Priya finds a co-host in ten minutes Collaborate

1 · Compass
Opens Collab Compass for "HackNSUT" and sees DTU Design Society at a 94% match.

2 · Trust
Reads the reasons, the 4.8 trust score and 11 past collabs.

3 · Request
Sends a co-host request with the date Date Doctor suggested.

4 · Accept
DTU accepts in the chat thread; a Collab Room and Charter are created instantly.

5 · Launch
The joint event goes live and 42 nearby colleges are reached through targeting and syndication.

J2 · Aarav discovers life beyond his campus Discover

1 · Onboard
Picks interests and verifies with his college email OTP (Tier 1).

2 · Pulse
Sees 12 events within 8 km this weekend on Campus Pulse.

3 · Follow
Follows 3 communities, and his feed and digest start to personalise.

4 · Register
Registers in one tap with Campus Passport and gets a QR.

5 · Proof
Checks in, earns a badge and a verifiable certificate, and shares it on LinkedIn.

J3 · Dr. Mehta governs without friction College HQ

1 · Submit
A club submits an event with external guests and a ₹200 fee.

2 · Risk flags
HQ highlights the outsiders, the fee and the late end time, and suggests a checklist.

3 · Approve
One tap approves it, the event publishes and an audit log entry is written.

4 · Registry
The completed collab is auto-logged in the college's collaboration registry.

5 · Report
At semester end, the activity report is exported as PDF or CSV.

J4 · The yearly handover that never loses anything Sustain

1 · Term ends
HQ prompts the club to start succession 30 days before term end.

2 · Nominate
The outgoing president nominates successors; the faculty coordinator approves.

3 · Transfer
Roles, admin rights and social connectors move atomically to the new team.

4 · Vault
Playbooks, the sponsor CRM and partner history are carried over in full.

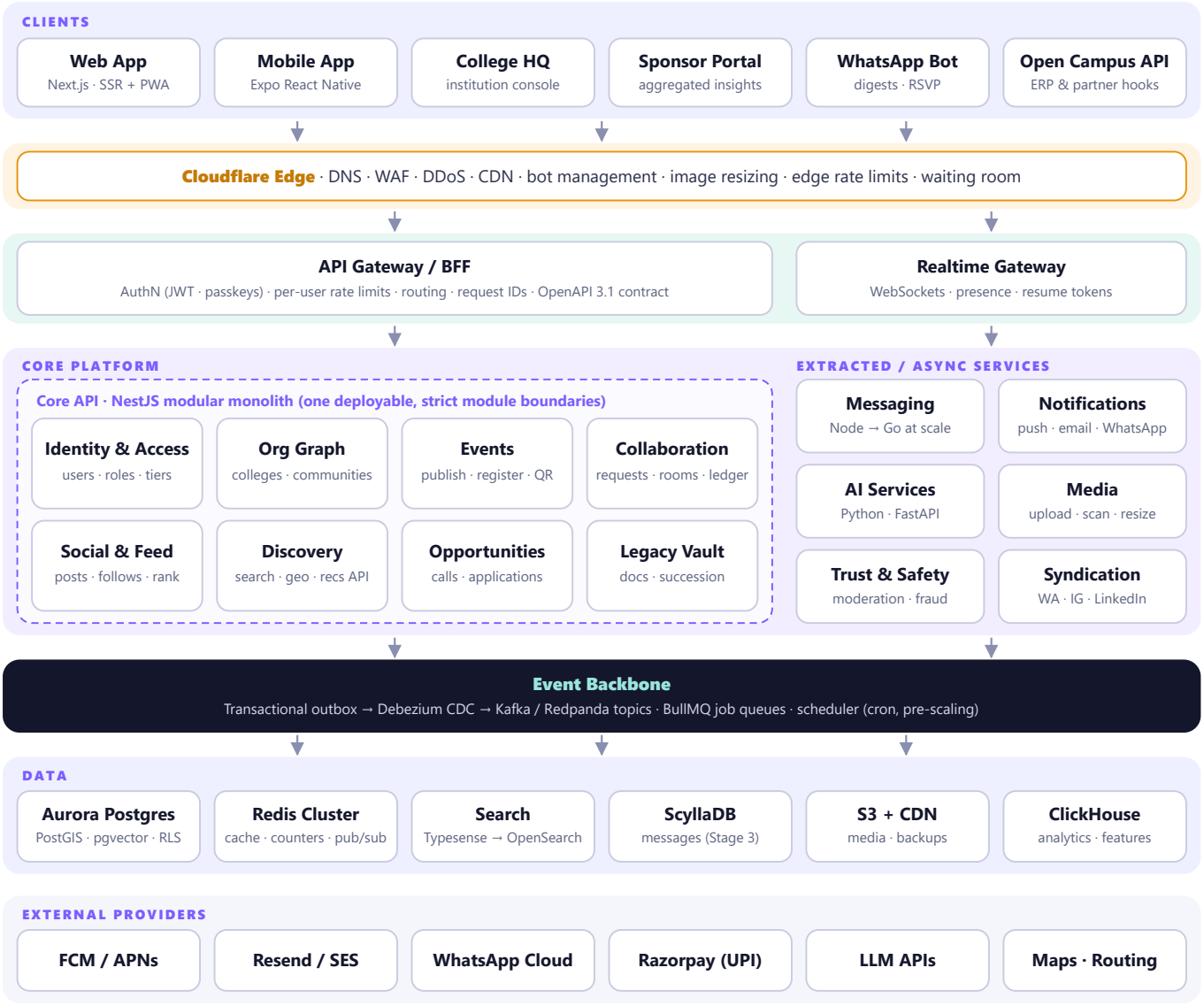
5 · Ask
The new lead asks the Vault "who sponsored us last year?" and gets cited answers.

> **Instrumented by default.** Each journey step emits a product event (for example `collab.request.sent` or `passport.checkin`) into ClickHouse, which gives us funnel dashboards from day one and training data for the ranking models in §12.

SYSTEM ARCHITECTURE

07 A platform designed to *grow without rewrites*

We start with one well-structured deployable, connect everything through events and push all heavy work off the request path. Services split out only when load, team size or failure isolation calls for it.



Cross-cutting: OpenTelemetry tracing · Sentry · feature flags · KMS encryption · Secrets Manager · least-privilege IAM · audit log

Figure 3 · Logical architecture. Solid boxes are deployables; the dashed group is one deployable with internal module boundaries.

Architecture principles

1 · Modular monolith first

One codebase and one deploy for the core, with modules that never touch each other's tables directly (enforced with lint rules and separate schemas). Extracting a module later is a deployment change, not a rewrite.

2 · Postgres is the truth; everything else is a projection

Search indexes, feeds, counters and analytics are derived from domain events and can be rebuilt by replaying. That kills a whole class of consistency bugs.

3 · Event-driven by default

Every state change publishes a versioned domain event through a transactional outbox. Consumers are idempotent, and nothing slow (emails, AI, indexing) sits on the request path.

4 · Design for spikes, not averages

Fest launches and registration openings create 10–50× bursts. We use predictive pre-scaling, a waiting room, atomic seat counters and load shedding (§15).

5 · Privacy & safety by design

DPDP-aligned consent, data minimisation, minors' protections and India data residency are part of the schema and the defaults, not a later audit fix.

6 · AI with guardrails

Structured outputs validated against schemas, human confirmation before anything publishes, per-tenant cost quotas and a non-AI fallback for every AI feature.

Request lifecycle: "publish an event"

Edge
Cloudflare WAF and rate limit; the request is routed to the nearest healthy origin.

›

Gateway
JWT verified, trace ID attached, idempotency key checked in Redis.

›

Core API
Policy check (event:publish), Zod validation, then a single transaction writes the `events` row and the `outbox` row.

›

CDC
Debezium streams the outbox row to Kafka topic `event.lifecycle`.

›

Consumers
Search index, feed fan-out, reach-targeted notifications, syndication and the Date Doctor index all update in parallel.

Client-perceived latency: about 120 ms (one DB transaction). Everything else completes asynchronously within seconds, and the event is searchable in under 5 s (p95).

SERVICES & TECHNOLOGY

08 Bounded contexts and the *stack behind them*

SERVICE / MODULE	RESPONSIBILITIES	OWNS DATA	LOAD PROFILE	RUNTIME
Identity & Access	Sign-in, sessions, verification tiers, RBAC/ABAC policies, device list	users, credentials, role_grants	Read-heavy, latency-critical	Core API
Org Graph	Colleges, communities, memberships with terms, follows, succession	colleges, communities, memberships, follows	Read-heavy, cacheable	Core API
Events	Create/publish, reach, registration, seats, QR check-in, certificates	events, event_hosts, registrations	Spiky (launches)	Core API + Redis
Collaboration	Requests, rooms, tasks, ledger, charter, resources, reviews, credits	collab_*, resources, reviews	Moderate, realtime	Core API
Social & Feed	Posts, comments, reactions, timelines, ranking	posts, reactions, timelines (Redis)	Read-heavy	Core API + Redis
Discovery	Search API, geo queries, recommendation serving, Ask CampusConnect	search index (derived)	Read-heavy	Core API + Typesense
Messaging	DMs, community channels, collab threads, receipts, presence	conversations, messages	Write-heavy, realtime	Node → Go; Scylla
Notifications	In-app, push, email, WhatsApp, digests, quiet hours, preferences	notifications, preferences	Bursty, async	Workers
AI Services	Embeddings, poster extraction, Compass scoring, RAG, moderation models	embeddings, prompt registry	API/GPU-bound, async	Python FastAPI
Media	Presigned upload, malware scan, resize/transcode, CDN URLs	object metadata	Bandwidth-heavy	S3 + workers
Trust & Safety	Reports, review queue, spam & fraud signals, appeals, bans	reports, cases, signals	Async	Python + rules
College HQ & Analytics	Approvals, dashboards, exports, sponsor insights (k-anonymous)	approvals, aggregates	Batch / OLAP	Core API + ClickHouse

Technology stack and why

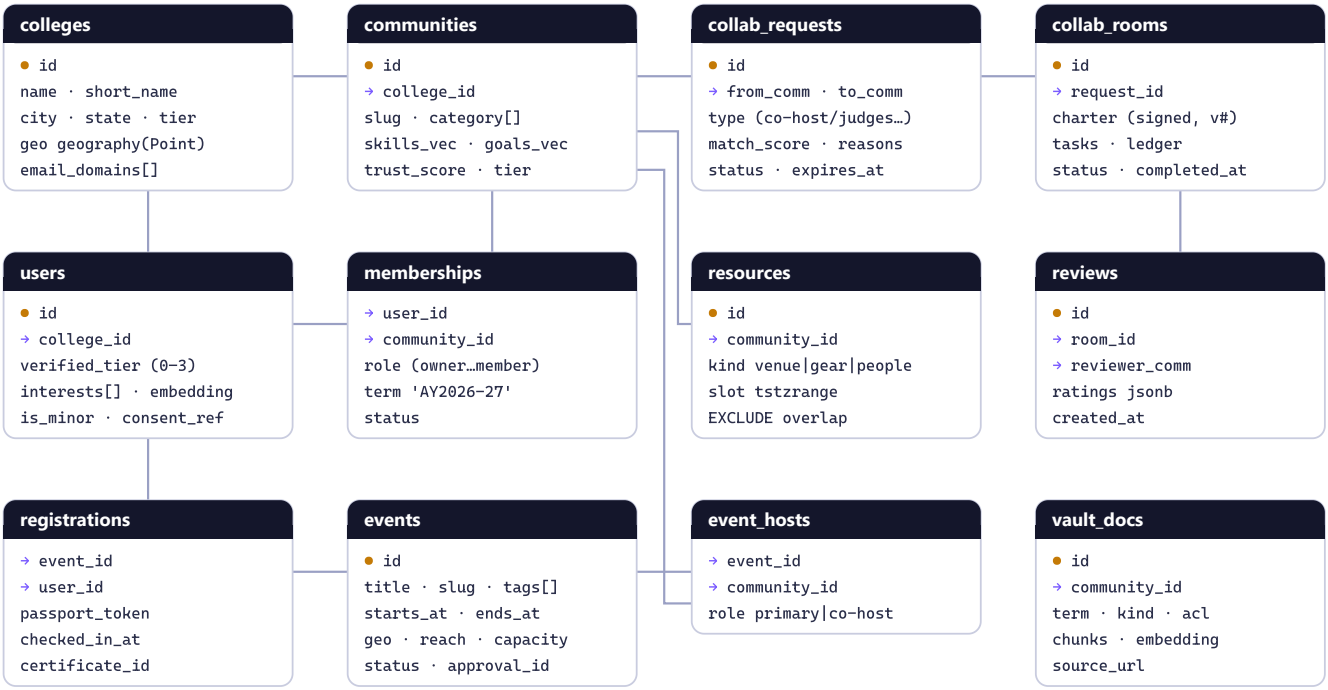
LAYER	CHOICE	WHY	CONSIDERED
Monorepo	Turborepo + pnpm	Web, mobile and API share Zod schemas, the API client and design tokens	Nx
Web	Next.js 15 (React 19, RSC, ISR)	SEO for public community/event pages; ISR absorbs launch spikes; PWA	Remix, SvelteKit
Mobile	Expo React Native + Expo Router + EAS	One codebase for iOS and Android, OTA updates, push and deep links built in	Flutter
Client state	TanStack Query + Zustand + MMKV	Cache-first, offline persistence, optimistic updates	Redux Toolkit
API	NestJS (TypeScript), REST + OpenAPI 3.1	Modules, DI and guards map cleanly to bounded contexts and RBAC	Fastify, tRPC
ORM / validation	Drizzle + Zod	SQL-first and type-safe; one schema from form to database	Prisma

Database	Aurora PostgreSQL 16 + PostGIS + pgvector	Relational + geo + vectors in one engine; Global Database for DR	MongoDB
Cache / queues	Redis (ElastiCache) + BullMQ	Sub-ms reads, atomic counters, pub/sub, delayed jobs	Memcached, SQS
Streaming	Kafka (MSK) / Redpanda + Debezium	Durable, replayable domain events; CDC keeps writes transactional	RabbitMQ
Search	Typesense → OpenSearch at scale	Typo tolerance, facets, geo and vector re-rank	Algolia, Meilisearch
Analytics	ClickHouse + PostHog	Columnar OLAP for dashboards and ML features; product funnels and flags	BigQuery
AI	Python FastAPI, LLM APIs, open embedding model, LightGBM	Structured extraction, RAG, learning-to-rank; model-agnostic gateway	—
Infra	AWS EKS + Karpenter, Terraform, Argo CD	Autoscaling on spot + on-demand; GitOps; reproducible environments	ECS, GCP
Edge	Cloudflare (CDN, WAF, Waiting Room, R2)	Indian PoPs, DDoS protection, cheap egress	CloudFront
Observability	OpenTelemetry → Grafana (Prometheus, Loki, Tempo) + Sentry	Vendor-neutral traces, metrics and logs; mobile crash reporting	Datadog

DATA ARCHITECTURE

09 A graph modelled in *relational* form

Twelve core entities carry the whole product. Terms on memberships, trust on communities and slots on resources are the modelling choices that make the novel features possible.



• primary key · → foreign key · also: follows, posts, conversations, messages, notifications, opportunities, approvals, outbox, audit_log (omitted for clarity)

Figure 4 · Core entity model.

Polyglot persistence: the right store for each shape of data

DATA	STORE	WHY
Core relational (users, orgs, events, collabs)	PostgreSQL (Aurora)	ACID, constraints, joins, row-level security for college-scoped data
Geo (campuses, venues)	PostGIS	Radius and bounding-box queries on GIST indexes; powers Pulse and Date Doctor
Embeddings (communities, users, vault)	pgvector (HNSW)	One database at the start; moves to a dedicated vector index only if recall or latency demands it
Messages	Postgres monthly partitions → ScyllaDB	Append-heavy; partition key `conversation_id`, clustering by ULID
Hot reads, counters, timelines, sessions	Redis Cluster	Sub-millisecond reads; atomic seat counters via Lua
Search documents	Typesense / OpenSearch	Typo tolerance, facets, geo filters, synonyms, Hinglish transliteration
Product & ML events	ClickHouse	Billions of rows, cheap aggregation; feature generation for ranking
Media, exports, backups	S3 / R2 behind CDN	Durable, cheap; presigned direct uploads keep the API stateless

Database techniques that matter

```
-- Resource Exchange: double-booking is impossible, enforced by the database itself
CREATE EXTENSION btree_gist;
ALTER TABLE resource_bookings ADD CONSTRAINT no_overlap
  EXCLUDE USING gist (resource_id WITH =, slot WITH &&) WHERE (status = 'confirmed');

-- Campus Pulse / Date Doctor: published events within 8 km in a time window
SELECT id, title FROM events
WHERE status = 'published' AND starts_at BETWEEN $1 AND $2
  AND ST_DWithin(geo, ST_MakePoint($lng, $lat)::geography, 8000);
```

- ✓ **ULIDs** as primary keys: sortable by time, safe to generate on the client for offline creates.
- ✓ **Partitioning:** messages and notifications by month; old partitions are detached and archived to S3 (Parquet).
- ✓ **Denormalised counters** (member_count, interested_count) are updated from events and reconciled nightly, so hot rows never lock.
- ✓ **Zero-downtime migrations** follow expand → migrate → contract, verified in CI against a production-sized snapshot.

EVENT-DRIVEN BACKBONE

10 Every change, *reliably* everywhere

Writing to the database and publishing to Kafka in two separate steps loses events when one step fails. The transactional outbox makes the business write and the event a single atomic operation.

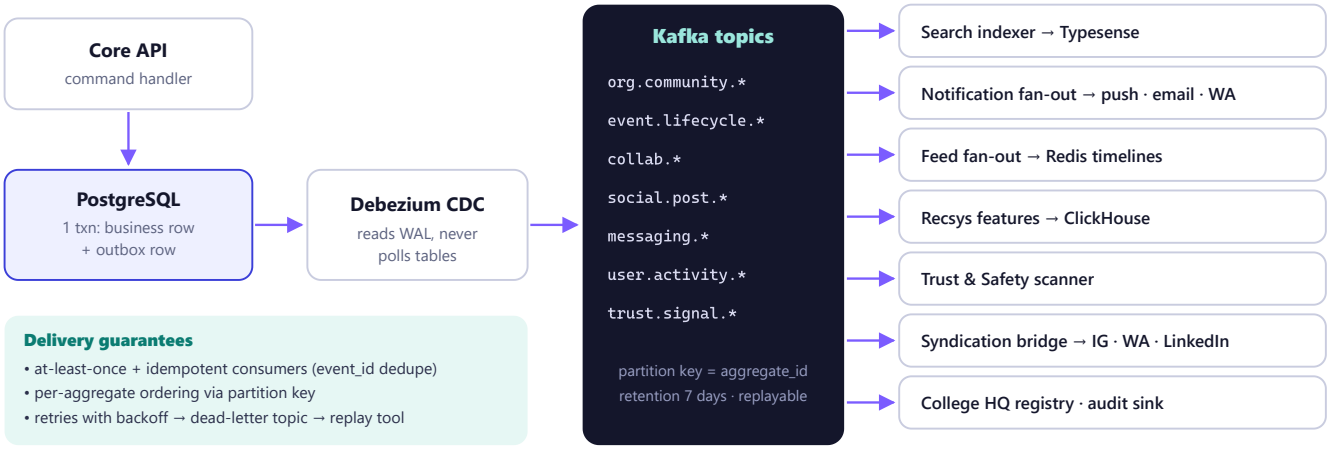


Figure 5 · Outbox → CDC → Kafka → independent consumers. Adding a feature means adding a consumer, never editing the write path.

Domain event catalogue (excerpt)

EVENT	MAIN CONSUMERS
CommunityVerified	Search, notifications, trust score
EventPublished	Search, feed, reach-targeted notify, syndication, Date Doctor
RegistrationCreated	Passport, analytics, organizer alerts
CollabRequested	Notifications, messaging (open thread)
CollabAccepted	Room provisioning, charter draft, HQ registry
CollabCompleted	Review prompts, credits, Compass training data
MessageSent	Offline push, async moderation, analytics
TermHandoverApproved	Permission transfer, vault ACLs, notifications

Versioned event envelope

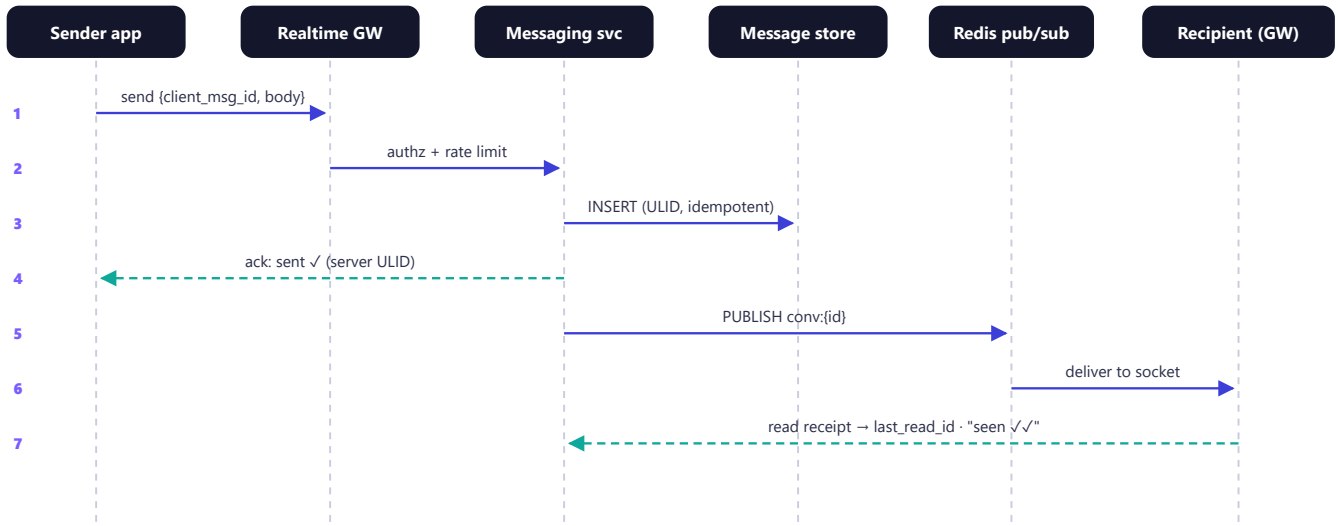
```
{
  "event_id":     "01JB7Z3K9X...",   // ULID, dedupe key
  "type":         "EventPublished",
  "version":      2,                 // schema registry
  "aggregate_id": "evt_8f2...",      // partition key
  "occurred_at":  "2026-11-02T10:15:04Z",
  "actor":        "usr_41a...",
  "trace_id":     "4bf92f35...",     // OTel propagation
  "payload": { "reach": "city", "geo": [77.03, 28.61],
               "hosts": ["com_nsut_code",
"com_dtu_design"] }
}
```

Schemas are stored in a registry, and backward-compatible changes are enforced in CI, so producers and consumers can deploy independently.

REAL-TIME MESSAGING

11 Chat that feels *instant* and never loses a message

Messaging drives the most writes, connections and safety risk in the system, so it's the first service we extract and the one we design most carefully.



Recipient offline?
MessageSent → Notification svc sends a push (collapsed per conversation, quiet hours respected). On reconnect the client syncs with GET /messages?after=<cursor>.

Figure 6 · Send path. The sender is acknowledged after a durable write, before fan-out.

Semantics

- At-least-once delivery, made exactly-once to the user with `client_msg_id` idempotency
- Ordering per conversation by server ULID
- Edits and deletes are events; tombstones are kept for sync

Conversation types

- 1:1 DM (Tier 1+, or a shared community)
- Community channels (announcements, general)
- Collab threads, linked to rooms
- Event Q&A, organizer-moderated

Scale tactics

- Stateless gateways: reconnect anywhere with a resume token, no sticky sessions
- Channels with > 5k members switch from push to pull-on-open with unread counters
- Presence via Redis keys with a 30 s TTL heartbeat
- Go gateway at Stage 3, about 50k sockets per node

Safety inside chat. New accounts can't DM strangers. Per-conversation rate limits, link scanning, one-tap report with message context, async ML moderation and DMs to minors restricted to verified shared communities.

INTELLIGENCE LAYER

12 Matching, ranking and AI that *explains itself*

Collab Compass scoring

$$\text{Match}(A, B) = 0.30 \cdot \text{sim}(\text{goals}_A, \text{goals}_B) + 0.25 \cdot \text{comp}(\text{skills}_A, \text{skills}_B) + 0.15 \cdot \text{geo}(d_{AB}) + 0.15 \cdot \text{trust}(B) + 0.10 \cdot \text{activity}(B) + 0.05 \cdot \text{novelty}(A, B) - \text{clash}(A, B)$$

comp = relevance-weighted $(1 - \text{sim}(\text{skills}))$ · $\text{geo}(d) = e^{-d/15\ \text{km}}$ · novelty rewards first-time cross-college edges · clash penalises calendar conflicts

The starting weights are hand-tuned. Once roughly 500 collaborations have completed, we replace them with a learning-to-rank model (LightGBM, LambdaMART) trained on *requested → accepted → completed → well-reviewed* outcomes. Each component is shown to users as a reason chip, so the explanation is the model itself rather than a generated rationalisation.

Recommendation pipeline

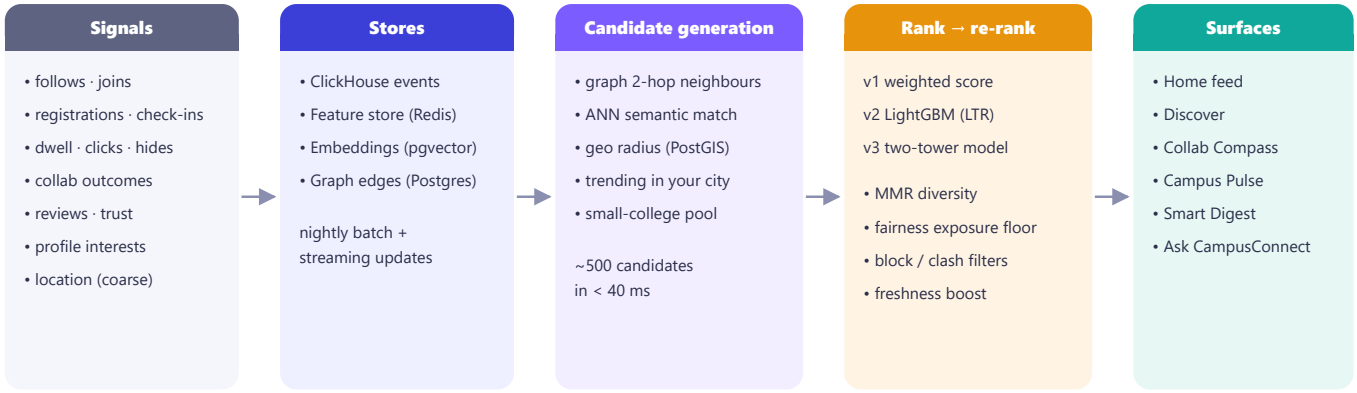


Figure 7 · One pipeline serves every discovery surface.

Fairness by design: "Small College Boost". Left alone, a network amplifies famous campuses. We reserve a minimum exposure share in every ranked list for communities from smaller and tier-2/3 colleges, and measure it as a first-class metric. This directly serves PS 2's goal of connecting *across* campuses.

AI services and guardrails

CAPABILITY	APPROACH	GUARDRAIL
Poster → event	Multimodal LLM with JSON-schema output, then geocoding	Per-field confidence; low-confidence fields require confirmation; manual form fallback
Ask the Vault (RAG)	Chunk + embed vault docs; retrieve top-k with ACL filter; cite sources	Only documents the asker can read; "I don't know" when retrieval is weak
Ask CampusConnect	LLM tool-calling over a whitelisted query API	No raw SQL; public data only; results rendered as cards
Moderation	Fast classifier first; LLM only for borderline cases; image NSFW model	Human review queue; appeal flow; audit trail
Translation	LLM translation of event descriptions (EN ↔ HI + regional)	Organizer can edit; original always available

Cost control: a model gateway routes small tasks to cheap models, embeddings are cached by content hash, jobs are batched off-peak, and per-community daily quotas apply. Prompt-injection hardening treats all user content strictly as data.

SEARCH & FEED

13 Find anything; see what *matters*

Hybrid search

- **Indexes:** communities, events, colleges, opportunities, people (opt-in only).
- **Retrieval:** BM25 keyword + vector semantic re-rank + geo filter + facets (category, date, reach, verified, fee).
- **India-aware:** synonyms ("E-Cell" ↔ entrepreneurship, "hack" ↔ hackathon), Hinglish transliteration, abbreviation expansion (NSUT, IGDTUW).
- **Freshness:** indexed from domain events; p95 under 5 s from write to searchable.
- **Zero-result rescue:** relax filters automatically, then suggest Ask CampusConnect.

Feed architecture

- **Stage 1, fan-out on read:** gather recent posts from follows, rank and cache for 60 s per user in Redis.
- **Stage 2, hybrid:** fan-out on write into Redis timelines for normal accounts; large accounts (> 10k followers) are merged at read time.
- **Mixing:** followed content, recommended communities, nearby events and opportunities, with diversity rules (max 2 consecutive items from one source).
- **Negative feedback:** "show less" signals feed straight into the ranker.

Feed ranking function (v1)

$$\text{score}(u, p) = w_1 \cdot \text{affinity}(u, \text{author}) + w_2 \cdot \text{interest}(u, p) + w_3 \cdot \text{proximity}(u, p) + w_4 \cdot \text{velocity}(p) + w_5 \cdot e^{-\Delta t/\tau} - \text{penalty}(\text{seen}, \text{hidden}, \text{reported})$$

Affinity is built from interactions with a decay. Velocity is engagement per hour, normalised by audience size so small clubs can trend. τ is around 18 h for posts and is tied to the start time for events.

SEO as a discovery channel

> Every public community, event and opportunity page is server-rendered with **schema.org JSON-LD** (`Event`, `Organization`). Events can then appear in Google's rich results, so a student searching "hackathon in Delhi this weekend" can land on CampusConnect without ever having heard of it. That's discovery the problem statement asks for, at zero marginal cost.

TRUST, SAFETY, SECURITY & PRIVACY

14 Safe for students, *credible* for colleges

Verification tiers

TIER	HOW IT'S EARNED	WHAT IT UNLOCKS
0 **Unverified**	Email or Google sign-in	Browse, follow, register for public events
1 **Student-verified**	College-domain email OTP, or ID upload with manual review	Join communities, DM members, apply to opportunities
2 **Community-verified**	Faculty coordinator approves via College HQ	City/national reach, collab requests, Resource Exchange listings
3 **Institution-verified**	Official domain + signed authorisation letter	College HQ, community verification, academic calendar import

Moderation pipeline

- Pre-publish classifier on text and images: allow, hold for review, or block.
- User reports with context; a reviewer queue with SLAs (P0 within 1 h, others within 24 h).
- Appeals, repeat-offender scoring, device-level ban evasion checks.
- Graph anomaly detection for fake engagement, review rings and spam collab requests.
- Link unfurling through a sandboxed fetcher with phishing and malware lists.

Protecting minors

- Users under 18 are flagged at onboarding and need verifiable parental consent, as the DPDP Act requires for children's data.
- Private profiles by default; excluded from people search and from all sponsor targeting.
- No behavioural tracking for ads; DMs only within verified shared communities.
- Event approvals flag overnight or off-campus events involving minors.

Security controls

AREA	CONTROLS
Identity	OIDC (Google), email OTP, passkeys (WebAuthn); 15-min access JWT + rotating refresh tokens with reuse detection; device/session list with remote logout
Authorisation	RBAC with community-scoped roles + ABAC policies (tier, college, minor status); Postgres row-level security as defence in depth; every privileged action audited
Data protection	TLS 1.3 everywhere; AES-256 at rest via KMS; field-level encryption for phone and date of birth; signed, expiring media URLs
Application	OWASP ASVS L2; Zod validation on every input; strict CSP; SSRF guards; presigned uploads with MIME sniffing and malware scan; per-route rate limits
Supply chain	Dependabot, CodeQL SAST, container scanning, SBOM generation, signed images (cosign), pinned dependencies
Operations	Least-privilege IAM, Secrets Manager with rotation, break-glass access with alerts, DAST (OWASP ZAP) in CI, annual third-party pentest, responsible-disclosure programme

DPDP Act 2023 alignment. Granular, purpose-specific and withdrawable consent (notices in English and Hindi). Self-service rights to access, correction and erasure, plus a grievance officer. Data minimisation and a published retention schedule. A breach-notification workflow. Verifiable parental consent for children. All personal data stored and processed in Indian AWS regions.

SCALABILITY ENGINEERING

15 Sized with numbers, *not adjectives*

Year-3 target: 5M registered users, 1M DAU, 3,000 colleges, 60,000 communities. Peaks are modelled at 5–10× the average because college traffic is extremely bursty.

METRIC	ASSUMPTION	AVERAGE	PEAK / TOTAL
API requests	60 requests / DAU / day	≈ 700 rps	≈ 5,000 rps
Messages	15 / DAU / day	≈ 175 msg/s	≈ 1,500 msg/s
Concurrent sockets	15% of DAU online at peak	—	≈ 150,000 connections
Notifications	8 / DAU / day	≈ 95 / s	≈ 2,000 / s (digest bursts)
Search	5 queries / DAU / day	≈ 60 qps	≈ 600 qps
Message storage	0.5 KB × 15M / day	≈ 7.5 GB / day	≈ 2.7 TB / year (90 days hot)
Media	60k communities × 30 images × 300 KB	≈ 540 GB / month	CDN serves > 95% of reads

Evolution path: grow in stages, never rewrite

STAGE 0 · HACKATHON
Prove the loop
- Next.js + Supabase (Postgres, Auth, Realtime, Storage)
- Vercel hosting
- LLM API
- **Same schema** as production

STAGE 1 · 0–100K USERS
Modular monolith
- Core API + realtime + workers on EKS
- Aurora + 1 read replica
- Redis, BullMQ, Typesense
- Cloudflare in front

STAGE 2 · 100K–2M
Extract & stream
- Messaging, notifications, AI extracted
- Kafka + Debezium outbox
- PgBouncer, replicas, partitions
- ClickHouse, OpenSearch

STAGE 3 · 2M–20M
Cells & shards
- Cell-based deployment by region
- ScyllaDB for messages
- Citus sharding by college_id
- Go realtime gateway; multi-region

Cell-based architecture (Stage 3): each "cell" is a full copy of the stack serving a set of regions (for example North, South, West, East), with a thin global directory routing users to their home cell. Cross-cell objects such as national events and collaborations live in a global tier. A failure is contained to one cell, and capacity grows by adding cells.

Surviving "Fest Rush"

Predictive pre-scaling
Registration opening times are known in advance, so the scheduler pre-scales API and realtime pods and warms caches 10 minutes before every launch above a reach threshold. We don't wait for autoscaling to react.

Virtual waiting room
For extreme launches, Cloudflare Waiting Room admits users at a sustainable rate with signed admission tokens. The rest see a fair queue position instead of errors.

Atomic seat inventory
Seats are decremented atomically in Redis (a Lua script), the registration is confirmed asynchronously in Postgres, and a nightly job reconciles the two. There's no overselling and no hot-row locks.

Load shedding & degradation
Priority classes are registration and messaging, then feed, then recommendations. Under pressure, recommendations fall back to cached "trending" and non-critical work is deferred to queues.

Caching layers

LAYER	WHAT	TTL	INVALIDATION
Edge (CDN)	Public pages (ISR), images, map tiles, OG cards	60 s – 1 day	Cache tags purged on domain events
Redis	Entity cache, feed pages, counters, sessions, rate limits	60 s – 5 min	Write-through + event-driven delete
In-process	Config, feature flags, policy rules	30 s	Pub/sub broadcast

Database guardrails: PgBouncer transaction pooling; reads routed to replicas where staleness is acceptable; a query budget (p95 under 20 ms) enforced with `pg_stat_statements` alerts; request coalescing on hot event pages.

DEPLOYMENT, RELIABILITY & DISASTER RECOVERY

16 Built to *stay up*

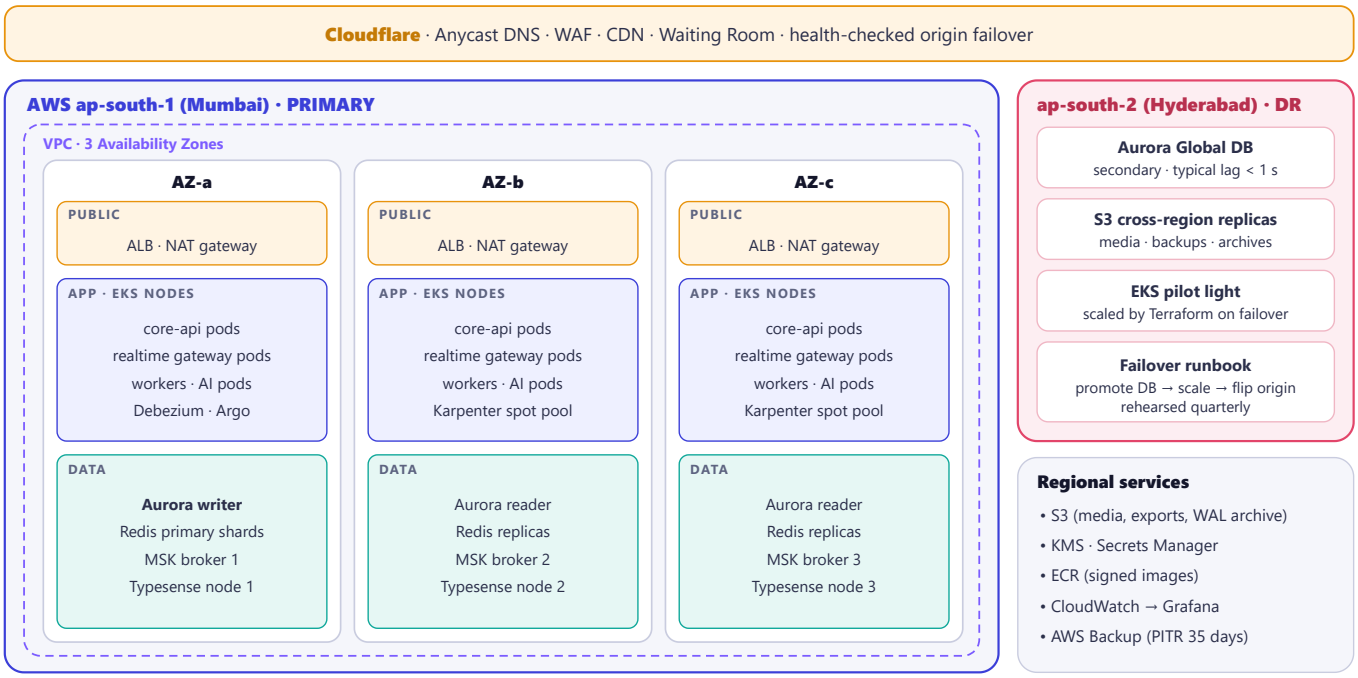


Figure 8 · Production topology: multi-AZ primary in Mumbai, warm DR in Hyderabad, all personal data inside India.

Service-level objectives

JOURNEY (SLI)	SLO
API availability	99.9% / month (43 min budget)
Read latency, p95	< 250 ms
Message delivery (online), p99	< 1 s
Registration success under load	99.95%
Notification handoff, p95	< 30 s
Search freshness, p95	< 5 s

Resilience

- ✓ **RPO < 1 min, RTO < 1 h** for a full-region loss; zero data loss for an AZ loss.
- ✓ Timeouts, retries with jitter, circuit breakers and bulkheads around every external API (WhatsApp, LLM, maps).
- ✓ Pod disruption budgets, topology spread across AZs, graceful shutdown that drains WebSockets.
- ✓ Error-budget policy: if the budget is burnt, feature releases freeze until reliability work lands.
- ✓ Quarterly game days: AZ kill, Redis failover, Kafka broker loss, region failover drill.

OBSERVABILITY, DEVOPS & QUALITY

17 Ship fast, *see everything*

Observability

- OpenTelemetry traces from client → gateway → service → Kafka → worker, linked by `trace_id`.
- RED dashboards per service, USE dashboards per resource.
- SLO burn-rate alerts (fast + slow windows) instead of noisy CPU alerts.
- Synthetic probes from Delhi, Mumbai, Bengaluru, Kolkata and Chennai.
- Sentry release health and crash-free sessions for mobile.
- On-call rotation, runbooks and blameless postmortems.

CI pipeline

- Trunk-based development; PRs gated on lint, type-check and affected-only builds (Turborepo cache).
- Unit (Vitest), contract tests against OpenAPI, integration (Testcontainers: Postgres, Redis, Kafka).
- E2E: Playwright (web), Maestro (mobile).
- Security: CodeQL, dependency and image scans, ZAP baseline.
- Preview environment per PR (Vercel + branch database).

CD & release

- GitOps with Argo CD; every environment declared in Git.
- Canary releases via Argo Rollouts, with automatic rollback on SLO regression.
- Feature flags for every user-facing change; kill switches for AI features.
- Mobile: EAS Build for store releases, EAS Update channels for staged OTA rollouts.
- DB migrations: expand → migrate → contract, never breaking N-1.

Environments

Local
Docker Compose: Postgres + PostGIS, Redis, Redpanda, Typesense, MinIO. Seeded data.

PR preview
Ephemeral stack per PR with anonymised fixtures; link posted on the PR.

Staging
Production-like, with load tests (k6) replaying a fest-rush profile weekly.

Production
Canary at 5%, then 25%, then 100%, gated on error rate and latency.

Quality bar

80%
coverage on domain modules (policies, scoring, ledger)

< 15 min
commit-to-production lead time

< 5%
change failure rate target

99.5%
crash-free mobile sessions

Repository layout

```
campusconnect/
├── apps/
│   ├── web/          # Next.js: public pages (SSR/ISR) + app + College HQ + sponsor portal
│   ├── mobile/       # Expo Router app (iOS + Android)
│   ├── api/          # NestJS modular monolith: identity, org, events, collab, social, discovery…
│   ├── realtime/     # WebSocket gateway
│   ├── worker/       # Kafka/BullMQ consumers, schedulers, pre-scaler
│   └── ai/           # Python FastAPI: extraction, embeddings, Compass, RAG, moderation
├── packages/
│   ├── schemas/      # Zod schemas = single source of truth for types
│   ├── api-client/   # generated from OpenAPI
│   ├── ui/           # design tokens + shared components
│   ├── db/           # Drizzle schema + migrations
│   └── events/       # domain event contracts + registry
└── infra/            # Terraform modules, Helm charts, Argo apps
```

CLIENT ARCHITECTURE

18 Web and app, *one brain*

The web and mobile apps share schemas, the API client, design tokens and translations, so a feature is designed once and delivered on both.

Web (Next.js)

- **React Server Components** for public pages: fast first paint, indexable, JSON-LD.
- **ISR** on event and community pages, so launch spikes hit the CDN instead of the origin.
- Client components for the logged-in app, using TanStack Query for cache and optimistic updates.
- PWA install and Web Push for students who won't install another app.
- College HQ and the sponsor portal are separate route groups with their own RBAC layouts.

Mobile (Expo React Native)

- **Offline-first:** query cache persisted to MMKV; writes queued in a local outbox and replayed on reconnect.
- Universal links and deep links for every shareable entity (event, community, collab, certificate).
- Push categories with inline actions (RSVP, accept collab, reply).
- Organizer mode: QR scanner verifies Passport tokens **offline** using a cached public key.
- Biometric re-auth for admin actions; OTA updates for fast fixes.

Built for every phone and every network

Lite mode

Under 150 KB of initial JS on public pages, text-first feed, adaptive image quality using the `Save-Data` hint and network type.

Languages

English and Hindi at launch, then Tamil, Telugu, Bengali and Marathi. ICU plurals, locale-aware dates, AI-assisted translations that people can edit.

Accessibility

WCAG 2.2 AA, 44 px touch targets, dynamic type, screen-reader labels, reduced-motion variants, colour-blind-safe status colours.

Performance budgets

METRIC	WEB TARGET	MOBILE TARGET
Largest Contentful Paint (4G)	< 2.0 s	—
Interaction to Next Paint	< 200 ms	—
Cold start (mid-range Android)	—	< 2.5 s
App download size	—	< 30 MB
Feed scroll	60 fps	60 fps (FlashList)

API DESIGN

19 A contract-first, *open* API

REST with an OpenAPI 3.1 contract, versioned under `/v1`. Clients are generated from the spec, so web, mobile and partners can never drift from it.

```
# Identity
POST /v1/auth/otp:request · /v1/auth/otp:verify
POST /v1/auth/refresh · /v1/auth/passkeys
POST /v1/me/verifications        # college email OTP

# Org graph
GET  /v1/colleges?city=&q=
GET  /v1/communities?near=lat,lng&radius=&cat=
POST /v1/communities/{id}/join
POST /v1/communities/{id}/succession

# Events
POST /v1/events:extract          # poster → draft
POST /v1/events · POST /v1/events/{id}:publish
GET  /v1/events/{id}/date-doctor?from=&to=
POST /v1/events/{id}/registrations
POST /v1/checkins                # Passport token
```

```
# Collaboration
GET  /v1/compass?for_event=&type=co-host
POST /v1/collabs · PATCH /v1/collabs/{id}
GET  /v1/rooms/{id} · POST /v1/rooms/{id}/tasks
POST /v1/rooms/{id}/charter:approve
GET  /v1/resources?kind=venue&near=
POST /v1/resources/{id}/bookings

# Social, messaging, discovery
GET  /v1/feed?cursor=
GET  /v1/conversations/{id}/messages?after=
GET  /v1/search?q=&type=&near=
POST /v1/ask                     # Ask CampusConnect
POST /v1/media:upload-url · POST /v1/reports
```

Conventions

- Cursor pagination everywhere
- `Idempotency-Key` on every write
- ETags + `If-None-Match`
- Errors as RFC 9457 problem+json
- Sparse fieldsets: `?fields=`

Rate limits

- Anonymous: 60 / min / IP
- Authenticated: 600 / min / user
- Auth endpoints: 5 / min / IP
- Messaging: 30 msgs / min / user
- Headers: `RateLimit-*` (IETF draft)

Open Campus API

- OAuth client-credentials for colleges and partners
- HMAC-signed webhooks: `event.published`, `registration.created`, `collab.completed`
- ERP/LMS sync of academic calendars
- Scoped, audited, revocable keys

GO-TO-MARKET, BUSINESS MODEL & METRICS

20 Solving the *empty-room* problem

The usual weakness of a network product is that it needs a network to be useful. Our launch strategy goes after that directly: seed, give single-player value, then let network loops take over.

1 · Seed the graph

Pre-create claimable pages for the top 500 colleges and their best-known communities in Delhi NCR from publicly available information. "Claim your page" goes through verification, so communities arrive to a page that already exists.

2 · Single-player value

Launchpad, Syndication, Legacy Vault and Date Doctor (on public calendars) are useful to *one* club with nobody else on the platform. That's the reason to sign up on day one.

3 · Network loops

Every shared event link brings in students. Every collab request invites a partner community. Every Passport certificate shared on LinkedIn advertises the platform.

THE PROMOTION GROWTH LOOP



Business model

WHO	WHAT THEY PAY FOR
Students	Nothing, ever
Communities	Free core; **Pro** plan for advanced analytics, unlimited rooms, custom branding
Colleges	**College HQ** annual licence: approvals, reports, calendar sync, branding
Sponsors & brands	**Sponsor Pool** deals and clearly labelled promoted opportunities; performance-based pricing
Paid events	Small platform fee on ticketing via Razorpay (UPI)

Guardrails: we never sell personal data, never target minors and always label promoted content.

Metrics that matter

North Star: cross-campus collaborations completed per month. It only goes up when discovery, trust, promotion and collaboration all work, which makes it the best single measure of PS 2.

- % of communities verified and weekly-active
- Share of event registrations coming from *other* colleges
- Collab request → acceptance → completion rates
- Time-to-first-collab for a new community
- Small-college exposure share (fairness)
- D30 retention of students; digest click-through rate

EXECUTION

21 Hackathon build plan, roadmap *& risks*

Build sprint (Stage 0)

DAY	BUILD	DONE WHEN
Day 1	Monorepo, Next.js + Supabase, full schema (§09), auth with college-email verification, RLS policies	Two users from different colleges can sign in with verified tiers
Day 2	Community pages, memberships and roles, Event Launchpad (poster → draft with an LLM), reach targeting	A poster becomes a published event in under 30 s
Day 3	Discovery: search, Campus Pulse map (PostGIS + MapLibre), Date Doctor clash strip	"Events within 8 km this weekend" works on seeded data
Day 4	Collab Compass scoring, collab requests, real-time chat thread, auto-created Collab Room with charter	The two-phone demo flow works live end-to-end
Day 5	Seed realistic data (20 colleges, 80 communities, 60 events), Passport QR, polish, PWA, demo video backup	3-minute demo rehearsed 5 times without a failure

3-minute demo script

0:00 · Hook
"Priya needs judges and a design partner. Today she has 3 cold DMs and no reply."

0:30 · Launch
Drop a poster; the event is live in 20 s.

1:00 · Date Doctor
It flags a clash with a nearby fest; she picks the suggested date.

1:20 · Compass
A 94% match with reasons; she sends a co-host request.

1:45 · Live accept
Second phone (DTU) accepts, and the Collab Room appears in real time.

2:15 · Reach
A student 6 km away sees it on Pulse and registers with Passport.

2:40 · Scale
HQ registry, then the architecture slide and close.

Demo safety net: seeded data, AI responses cached for the demo poster, an offline fallback for every external API and a recorded backup video.

Roadmap

0–3 months

Delhi NCR pilot

20 colleges, 300 communities. Core pillars, College HQ beta, WhatsApp digests, Hindi.

3–9 months

Five metros

Bengaluru, Mumbai, Pune, Hyderabad, Chennai. Paid College HQ, Resource Exchange, learning-to-rank Compass, Stage 2 infra.

9–18 months

National

Regional languages, Sponsor Pool, Open Campus API, ticketing, alumni mentors, Stage 3 cells.

Risks and mitigations

RISK	MITIGATION
Cold start / empty network	Claimable seeded pages, single-player value, city-by-city density, fest-season launch with ambassadors
Fake communities and spam	Verification tiers, faculty approval for reach, graph anomaly detection, rate limits
Minors' safety	Parental consent, private-by-default, DM restrictions, no sponsor targeting, fast moderation SLAs
AI errors and cost	Human confirmation, confidence thresholds, model routing, quotas, non-AI fallbacks
Third-party API policy changes (Meta)	Adapter pattern per channel, email and web push as fallbacks, no single-channel dependency
Leadership churn hurting adoption	Legacy Vault turns churn into a *reason* to stay; succession prompts driven by HQ

SUMMARY

22 Why CampusConnect *wins*

How the blueprint maps to what hackathon judges typically score.

CRITERION	HOW WE DELIVER
Problem understanding	Four stated pains broken into root causes, plus six hidden problems (churn, clashes, asymmetry, trust, institutional blind spot, wrong channel)
Innovation	Complementarity matching, clash-aware scheduling, Resource Exchange with database-enforced bookings, leadership-handover RAG, verifiable cross-campus Passport
Completeness	Every PS 2 sentence mapped to a feature and a metric; web, mobile, institution and sponsor surfaces
Technical depth	Outbox + CDC, polyglot persistence, hybrid search, learning-to-rank, real-time messaging semantics
Scalability	Numbers-based capacity plan, staged evolution to cell-based architecture, Fest Rush engineering
Reliability & security	SLOs with error budgets, multi-AZ + DR region, OWASP ASVS L2, DPDP alignment, minors' protections
Feasibility	The Stage 0 build uses the production schema; a 5-day plan with a scripted, fail-safe demo
Impact & viability	A North Star tied to the PS, fairness for small colleges, a revenue model that never charges students

THE ONE-LINE PITCH

CampusConnect helps student communities *find* the right partner, *build* something together and *remember* how they did it, across every campus in India.

Discover · Connect · Promote · Collaborate · Sustain

Glossary

CDC: change data capture; streaming database changes from the write-ahead log.
Outbox: a table written in the same transaction as business data, then published as events.
RLS: row-level security; Postgres-enforced access rules per row.
ISR: incremental static regeneration; cached pages refreshed in the background.
LTR: learning-to-rank; ML models that learn ordering from outcomes.
MMR: maximal marginal relevance; re-ranking for diversity.
RPO / RTO: how much data, and how much time, a disaster may cost.
DPDP: India's Digital Personal Data Protection Act, 2023.